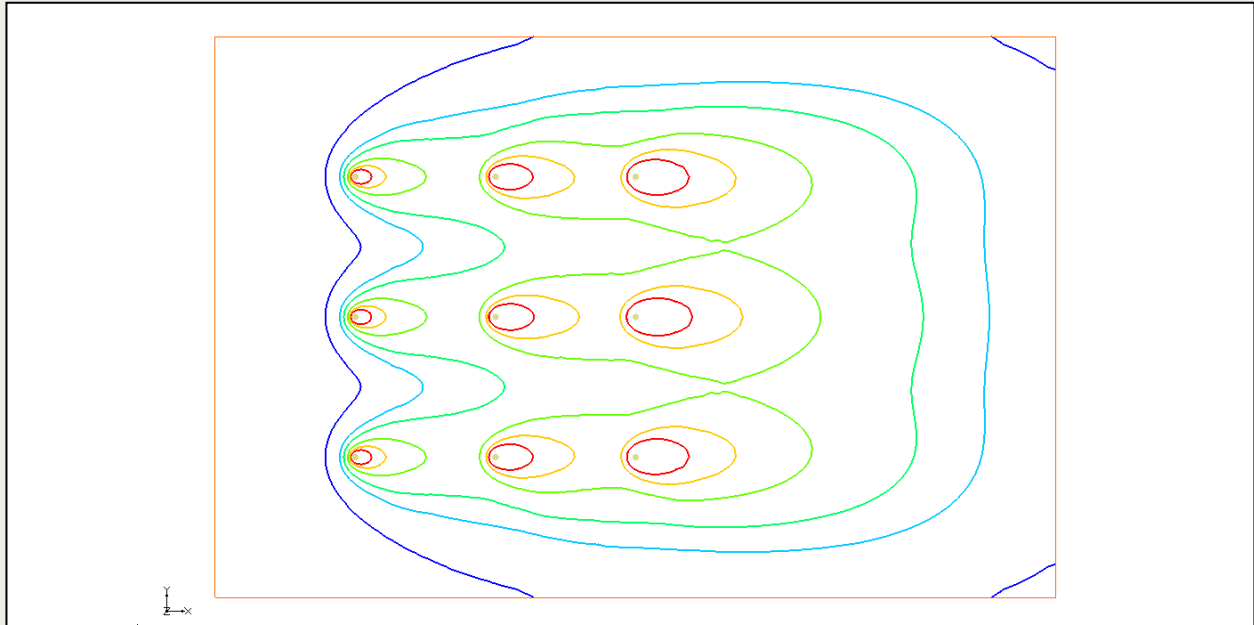


*GMS 10.9 Tutorial***MODFLOW 6 – Borehole Heat Exchangers**

Build a MODFLOW 6 simulation with Borehole Heat Exchangers using the Groundwater Energy (GWE) model

**Objectives**

This tutorial demonstrates creating a MODFLOW 6 Simulation using the Groundwater Energy (GWE) model. The simulation uses the solution of an existing flow model.

**Prerequisite Tutorials**

- Getting Started
- MODFLOW 6 Grid Approach

**Required Components**

- GMS Core
- MODFLOW-USG Model & Interface

**Time**

- 25–45 minutes

|           |                                      |           |
|-----------|--------------------------------------|-----------|
| <b>1</b>  | <b>Introduction .....</b>            | <b>2</b>  |
| <b>2</b>  | <b>Getting Started .....</b>         | <b>2</b>  |
| 2.1       | Opening the Existing Model .....     | 3         |
| <b>3</b>  | <b>Saving the Project.....</b>       | <b>3</b>  |
| <b>4</b>  | <b>Adding a GWE Simulation .....</b> | <b>4</b>  |
| <b>5</b>  | <b>Set TDIS Parameters .....</b>     | <b>4</b>  |
| <b>6</b>  | <b>Set IMS Parameters .....</b>      | <b>5</b>  |
| <b>7</b>  | <b>Set ADV Parameters .....</b>      | <b>5</b>  |
| <b>8</b>  | <b>Set CND Parameters .....</b>      | <b>5</b>  |
| <b>9</b>  | <b>Set ESL Parameters.....</b>       | <b>6</b>  |
| <b>10</b> | <b>Set EST Parameters.....</b>       | <b>6</b>  |
| <b>11</b> | <b>Set FMI Parameters .....</b>      | <b>7</b>  |
| <b>12</b> | <b>Set IC Parameters .....</b>       | <b>7</b>  |
| <b>13</b> | <b>Set OC Parameters .....</b>       | <b>7</b>  |
| <b>14</b> | <b>Set SSM Parameters .....</b>      | <b>7</b>  |
| <b>15</b> | <b>Saving the Simulation .....</b>   | <b>8</b>  |
| <b>16</b> | <b>Running MODFLOW 6 .....</b>       | <b>8</b>  |
| 16.1      | Rerun the flow simulation .....      | 9         |
| 16.2      | Run the energy simulation .....      | 9         |
| <b>17</b> | <b>Examine the Solution .....</b>    | <b>9</b>  |
| <b>18</b> | <b>Conclusion .....</b>              | <b>10</b> |

## 1 Introduction

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This tutorial illustrates how to model borehole heat exchangers with the Groundwater Energy (GWE) model. The tutorial is based on the example “Interacting Borehole Heat Exchangers in a Geothermal Setting” that is included with MODFLOW 6.

“Shallow groundwater geothermal investigations often include more than one borehole heat exchanger (BHE). In such applications, understanding the thermal interaction among multiple BHEs as well as on the flowing groundwater is made easier with a numerical groundwater flow and heat transport model. In this example, the accuracy of the groundwater energy transport (GWE) model is demonstrated for a convective-conductive porous domain with multiple thermally-interacting BHEs using an analytical solution...

“For this example, nine BHEs are arranged in a  $3 \times 3$  configuration with a spacing of 5 m from each other... The grid extent is 90 m $\times$  60 m. Each BHE represents a cylindrical source of heat with energy being added at a rate of 100 Wm using the energy source loading (ESL) package.”<sup>1</sup>

## 2 Getting Started

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Do as follows to get started:

1. If necessary, launch GMS.
2. If GMS is already running, select the *File* | **New** menu item to ensure that the program settings are restored to their default state.

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<sup>1</sup> MODFLOW 6 Development Team, with contributions from Chieh Ying Chen and Mike Toews, 2017, MODFLOW 6 – Example problems

## 2.1 Opening the Existing Model

Start with a previously created project.

1. Click the **Open**  macro to bring up the *Open* dialog.
2. From the *Files of type* drop-down, select “Project Files (\*.gpr)”.
3. Browse to the *mf6\_gwe\_geotherm\start* folder and select “start.gpr”.
4. Click the **Open** button to import the project and exit the *Open* dialog.

The project should be visible in the Graphics Window (Figure 1). The project contains a MODFLOW 6 flow simulation along with a 3D UGrid. A solution also exists for the flow model.

An energy simulation will be added to the project that will build on the flow simulation.

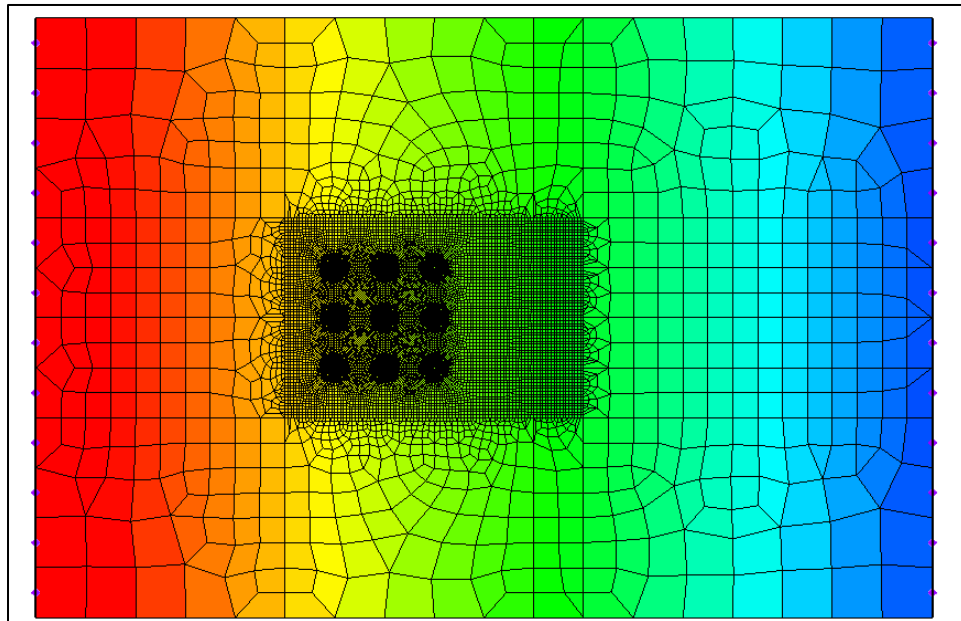


Figure 1: Initial project

## 3 Saving the Project

Before making any changes, save the project under a new name.

1. Select the *File* | **Save As...** menu item to bring up the *Save As* dialog.
2. From the *Save as type* drop-down, select “Project Files (\*.gpr)”.
3. For the *File name*, enter “geotherm.gpr”.
4. Click the **Save** button to save the project under the new name and close the *Save As* dialog.

It is recommended to periodically **Save**  while working through the tutorial and while working on any project.

## 4 Adding a GWE Simulation

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The next step is to add a GWE simulation.

1. In the Project Explorer, right-click the “ MODFLOW 6 Simulations” folder and select the **New Simulation** context menu item to bring up the *New MODFLOW 6 Simulation* dialog.
2. For the *Simulation name*, enter “gwe-geotherm”.
3. Under the *Models* section of the dialog, turn off the *GWF – Groundwater Flow Model* checkbox.
4. Turn on and select the *GWE – Groundwater Energy Transport Model* checkbox.
5. Under the *GWE – Groundwater Energy Transport Model* option, turn on the checkboxes for:
  - ADV – Advection
  - CND – Conduction and Dispersion
  - ESL – Energy Source Loading
  - EST – Energy Storage and Transfer
  - FMI – Flow Model Interface
  - SSM – Source and Sink Mixing
6. Click the **OK** button to close the *New MODFLOW 6 Simulation* dialog.

The new “ gwe-geotherm” simulation along with an “ energy” model will appear in the Project Explorer. Expand the simulation to reveal the packages inside.

## 5 Set TDIS Parameters

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The energy simulation will have fifty stress periods.

1. Under the “ gwe-geotherm” simulation, double-click on the “ TDIS” package to open the *Temporal Discretization (TDIS) Package* dialog.
2. In the only row in the *PERIODDATA* table, change the values to the following:
  - PERLEN: 1.0
  - NSTP: 12
  - TSMULT: 1.3
3. Click the **Add Rows...** button.
4. For the *Number of rows to add at bottom* field, enter “49”.
5. Click the **OK** button to exit the *Rows to Add* dialog.

The new stress periods are all one day long with one time-step. This is what is needed for this tutorial and no changes need to be made.

6. Click the **OK** button to exit the *Temporal Discretization (TDIS) Package* dialog.

## 6 Set IMS Parameters

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To set the Iterative Model Solution (IMS) parameters:

1. Under the “ gwe-geotherm” simulation, double-click on the “ IMS” package to open the *Iterative Model Simulation (IMS)* dialog.

The IMS package for the GWE model requires a lot of changes from the default values. In the interest of time, an existing package will be imported.

2. Click the **Import...** button to open the *Import* dialog.
3. Browse to the *mf6\_gwe\_geotherm\start* folder and select the “gwe-ims.ims” file.
4. Click the **Open** button to close the *Import* dialog.
5. Click the **OK** button at the message that the file has been imported successfully.
6. Click the **OK** button to close the *Iterative Model Simulation (IMS)* dialog.

## 7 Set ADV Parameters

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For the Advection (ADV) package, the scheme needs to be changed to “TVD”:

1. Under the “ gwe-geotherm” simulation, in the “ energy” folder, double-click on the “ ADV” package to open the *Advection (ADV) Package* dialog.
2. Under the *OPTIONS* section, set *SCHEME* to “TVD”.
3. Click the **OK** button to close the *Advection (ADV) Package* dialog.

## 8 Set CND Parameters

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Define the Conduction and Dispersion (CND) package properties:

1. Double-click on the “ CND” package to open the *Conduction and Dispersion (CND) Package* dialog.
2. In the *GRIDDATA* tab, select the *ALH* tab (ALH is the longitudinal dispersivity in the horizontal direction).
3. Turn on *Define* and verify the *Constant* value is “0.0”.
4. Select the *ATH1* tab (ATH1 is the transverse dispersivity in the horizontal direction).
5. Turn on the *Define* checkbox and verify the *Constant* value is “0.0”.
6. Select the *KTW* tab (KTW is the thermal conductivity of the simulated fluid).
7. Turn on the *Define* checkbox and set the *Constant* field to “48384.0”.
8. Select the *KTS* tab (KTS is the thermal conductivity of the solid aquifer material).
9. Turn on the *Define* checkbox and set the *Constant* field to “216000.0”.
10. Click the **OK** button to close the *Conduction and Dispersion (CND) Package* dialog.

## 9 Set ESL Parameters

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The Energy Source Loading (ESL) package is used to add energy to the nine cells representing the BHEs. The energy source loading rate (SENERRATE) increases for the first ten stress periods. In the interest of time, a CSV file will be imported containing the data for the first ten stress periods.

1. Double-click on the “ ESL” package to open the *Energy Source Loading (ESL) Package* dialog.
2. In the *Periods* section of the dialog, click the **Import CSV File**  button to open the *Import CSV File* dialog.
3. Browse to the *mf6\_gwe\_geotherm\start* folder.
4. Select the “*esl.csv*” file and click the **Open** button to close the *Import CSV File* dialog.
5. Change the *Period* to see the values for other stress periods.

Note: the values are the same at all nine cells, and the energy increases for the first ten stress periods, which are the only ones defined.

6. Click the **OK** button to close the *Energy Source Loading (ESL) Package* dialog.

## 10 Set EST Parameters

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Set up the Energy Storage and Transfer (EST) package parameters.

1. Double-click on the “ EST” package to open the *Energy Storage and Transfer (EST) Package* dialog.
2. In the *GRIDDATA* section, select the *POROSITY* tab.
3. Change the *Constant* value to “0.2”.
4. Select the *HEAT\_CAPACITY\_SOLID* tab.
5. Change the *Constant* value to “900.0”.
6. Select the *DENSITY\_SOLID* tab.
7. Set the *Constant* field to “2650.0”.
8. In the *Section* list, turn on the *OPTIONS* checkbox.
9. Under the *OPTIONS* section, turn on the *DENSITY\_WATER* checkbox and verify the value is 1000.
10. Turn on the *HEAT\_CAPACITY\_WATER* checkbox and for the value enter “4180.0”.
11. Turn on the *LATENT\_HEAT\_VAPORIZATION* checkbox and for the value enter “2500”.
12. Click the **OK** button to close the *Energy Storage and Transfer (EST) Package* dialog.

## 11 Set FMI Parameters

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The energy simulation will use the existing solution from the flow simulation. This is sometimes referred to as an “uncoupled” scenario. The FMI package is used to achieve this.

1. Double-click on the “ FMI” package to open the *Flow Model Interface (FMI) Package* dialog.
2. In the PACKAGEDATA section, turn on the “Let GMS determine the paths from the flow simulation when saving” checkbox.
3. Click the **Select Flow Simulation** button to open the *Select Flow Simulation* dialog.
4. Turn on the “gwf-geotherm” model checkbox.
5. Click the **OK** button to close the *Select Flow Simulation* dialog.
6. Click the **OK** button to close the *Flow Model Interface (FMI) Package* dialog.

## 12 Set IC Parameters

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The default Initial Conditions (IC) temperature in the model is 0.0 °C. No changes need to be made.

1. Double-click on the “ IC” package to open the *Initial Conditions (IC) Package* dialog.
2. In the GRIDDATA section, verify the *Constant* value is “0.0”.
3. Click the **OK** button to close the *Initial Conditions (IC) Package* dialog.

## 13 Set OC Parameters

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The default Output Control (OC) Option settings are to save the budget and temperature at every time step. No changes need to be made.

1. Double-click on the “ OC” package to open the *Output Control (OC) Dialog*.
2. In the PERIODS section, verify the *Preset output* value is “At every time step”.
3. In the spreadsheet, verify entries for *PRINT\_SAVE* include *RTYPE* entries for “BUDGET” and “TEMPERATURE”.
4. Click the **OK** button to close the *Output Control (OC) Dialog*.

## 14 Set SSM Parameters

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The Source and Sink Mixing (SSM) package parameters will make use of the parameters in the constant-head packages in the flow model.

1. Double-click on the “ SSM” package to open the *Source and Sink Mixing (SSM) Package* dialog.

Rows for the SSM packages could be added manually, but it’s easier to let GMS do it.

2. Under the *SOURCES* section, click the **Set Up From Flow Model** button to open the *Select GWF6* dialog.
3. Turn on the “gwf-geotherm” flow model checkbox.
4. Click the **OK** button to close the *Select GWF6* dialog.

GMS found the TEMPERATURE auxiliary variables in the two CHD packages and added them as two new rows in the table.

5. Click the **OK** button to close the *Source and Sink Mixing (SSM) Package* dialog.

## 15 Saving the Simulation

Before running the model simulation, the project and model native files need to be saved.

1. Click the **Save**  macro to save the project.

The project file has now been saved. However, the simulation files needed to run MODFLOW 6 have not yet been exported. To export these files, do the following:

2. In the Project Explorer, right-click on “ gwf-geotherm” and select the **Save Simulation** context menu item.

The files for the simulation have now been exported.

## 16 Running MODFLOW 6

It is now possible to run MODFLOW:

1. Right-click on “ gwf-geotherm” and select **Run Simulation**.

The *Simulation Run Queue* shows the output from MODFLOW 6. The report should show the following errors:

```

ERROR REPORT:

1. Could not find file ..\gwf-geotherm\gwf-geotherm_output\gwf-
geotherm.cbc

2. Could not find file ..\gwf-geotherm\gwf-geotherm_output\gwf-
geotherm.hds

UNIT ERROR REPORT:

1. ERROR OCCURRED WHILE READING FILE 'energy.fmi'

```

The errors indicate that the flow solution files are not in the location indicated by the FMI package. Recall that the FMI package was set up by using the option “Let GMS determine the paths from the flow simulation when saving”. That option assumes that all simulations in the project have solutions below a common subfolder named using the project name followed by “\_model”, so “geotherm\_models” in our case. However, the original flow solution is below the “start\_models” folder because the project was originally named “start.gpr”.



There are two ways to fix this problem:

1. Manually fix the paths in the FMI package to point back to the original solution files in the “start\_models” folder
2. Rerun the flow simulation to generate a new flow solution below the “geotherm\_models” folder

In this case, since the flow simulation runs quickly, option 2 will be used. This also has the advantage of keeping all the simulations and solutions together in one folder. If the flow simulation could not be rerun for some reason, option 1 would need to be used.

## 16.1 Rerun the flow simulation

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1. Click the **Remove** button in the *Simulation Run Queue* dialog.
2. Click the **Close** button to close the *Simulation Run Queue* dialog.
3. Right-click on the “gwf-geotherm” simulation and select the **Save Simulation** context menu item.
4. Right-click on the “gwf-geotherm” simulation again and select the **Run Simulation** context menu item.
5. If a dialog appears saying that the existing solution will be unloaded, click the **OK** button.
6. When MODFLOW 6 finishes, click the **Load Solution** button.

## 16.2 Run the energy simulation

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Now the flow solution exists in the expected location.

1. Right-click on the “gwe-geotherm” simulation and select the **Run Simulation** context menu item.
2. If a dialog appears saying that the existing solution will be unloaded, click the **OK** button.

The energy simulation will take several seconds to complete. This time there should be no reported errors.

3. When MODFLOW 6 finishes, click the **Load Solution** button.
4. Click **Close** to exit the *Simulation Run Queue* dialog.

## 17 Examine the Solution

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Review the solution by doing the following:

1. Under the “ gwe-geotherm” simulation, “ Solution” folder, expand the “ energy” folder and select the “ Temperature” dataset.
2. Change time steps using the *Up arrow* or *Down arrow* keyboard keys, or *Page Up* or *Page Down* keyboard keys. See how the contours show the temperature changing over time, refer Figure 2.

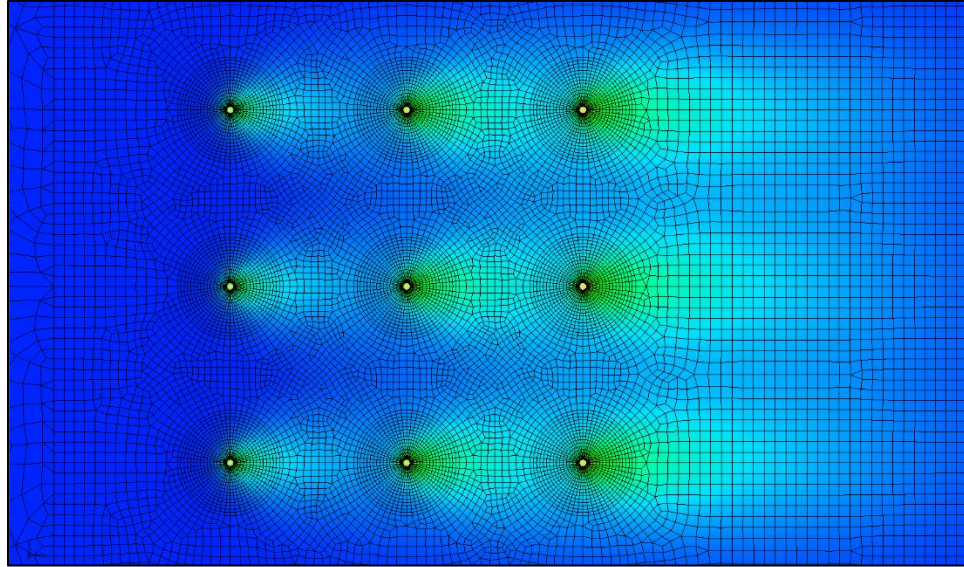


Figure 2: Temperature solution at final time step

## 18 Conclusion

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This concludes the “MODFLOW 6 – Groundwater Energy model with an Existing Flow Solution” tutorial. The following topics were discussed and demonstrated:

- A Groundwater Energy (GWE) simulation can be made to use an existing flow solution.
- The FMI package must contain the correct paths to the flow solution files.